Get Fit with MA? Estimating the Impact of Medicare Advantage on Physical Activity

Jong Min Jung* Jing Li†

Abstract

Medicare Advantage (MA) plans increasingly offer fitness benefits, yet the evidence on their behavioral effects remains limited. This paper studies whether MA enrollment is associated with a differential change in physical activity at the Medicare eligibility threshold. Using a difference-in-discontinuities design and nationally representative survey data on workout behavior and insurance coverage, we find that MA enrollment is associated with an approximately 4 percentage point increase in the probability of daily exercise relative to non-MA beneficiaries. Estimates for ages below 65 show no evidence of differential pre-trends, while the post-65 pattern indicates that the effect is concentrated in the short run. The response is larger among women, non-White, and individuals without a college degree. Robustness checks, including placebo tests and a donut-hole analysis yield consistent results. These findings suggest that insurance design may affect preventive behavior and highlight supplemental benefits as an important policy margin for promoting healthy aging.

Keywords: Medicare Advantage, physical activity, fitness benefits, fuzzy regression discontinuity

*Department of Economics, University of Washington
†Department of Pharmacy, University of Washington

1 Introduction

As the U.S. population ages, promoting healthy behaviors among seniors has become both a public health imperative and nan economic necessity. Medicare accounts for 21% national health expenditure and 10% of federal budget (Cubanski and Neuman (2023)), and these financial burdens will intensify as adults over 65 outnumber children under 18 for the first time by 2034 (U.S. Census Bureau (2018)). At the same time, even modest improvements in health behaviors at older ages can yield substantial returns in longevity and quality of life, ultimately lowering health care utilization (Fries et al. (2011)). Among preventive behaviors, physical activity stands out as it is one of the most cost-effective interventions to promote healthy aging but it remains severely underutilized.

Physical inactivity is pervasive across all ages worldwide, particularly among older adults. In the United States, more than 25 percent of adults aged 65 and older report no leisure-time physical activity, and only 14 percent meet both the aerobic and muscle-strengthening components of the 2018 Federal Physical Activity Guidelines (CDC (2022)).These guidelines recommend at least 150 minutes of moderate-intensity or 75 minutes of vigorous-intensity exercise per week, along with muscle-strengthening activities on two or more days. Compliance varies significantly across demographic groups and regions : 17 percent of older adults in the West meet the combined standard compared with just 12 percent in the South, and seniors in metropolitan area are considerably more active than those in non-metropolitan areas(Elgaddal and Kramarow (2024)).

The health outcomes from physical inactivity among seniors are severe. Insufficient physical activity increases the risk of cardiovascular disease, diabetes, cognitive decline, musculoskeletal disorders (Warburton et al. (2006);Lee et al. (2012)), and is a leading contributor to falls which is the primary cause of injury-related death among seniors (Jiang and Wang (2022)). The economic burden is equally substantial as Medicare expenditures attributable to insuffi-

cient physical activity have been estimated at $117 billion annually in the U.S.(Carlson et al. (2015)). These figures highlight that physical inactivity is not merely a clinical concern but a first-order fiscal challenge for the Medicare program.

Physical inactivity also reflects and reinforces broader health inequalities. Structural barriers such as financial constraints, limited access to recreational facilities, lack of transportation, and insufficient social support disproportionately affect low-income, ethnic minorities, and less-educated populations(Elgaddal and Kramarow (2024)). These disparities imply that the populations most likely to benefit from increased physical activity are those least likely to engage in it. This pattern raises an important policy question about whether targeted interventions embedded in insurance benefit designs may reduce structural barriers to exercise and narrow disparities in physical activity among seniors.

Medicare Advantage(MA), the privately administered alternative to Traditional Medicare(TM), offers one promising lever. Unlike TM, most MA plans provide supplemental free fitness benefits such as SilverSneakers, Renew Active, Silver&Fit. These fitness programs provide free or subsidized gym memberships, instructor-led group exercise classes, and community-based fitness opportunities, which directly target the financial and logistical barriers to exercise among older adults. The availability of these benefits has grown rapidly. By 2025, about 95 percent of individual MA plans offer a free fitness benefit, and MA enrollment had risen to cover more than half of all Medicare beneficiaries, with that share projected to increase to 64 percent by 2033 (KFF (2025)). Nevertheless, rigorous evidence on whether MA fitness benefits change exercise behavior remains limited.

Existing evidence on the behavioral impact of MA fitness programs is largely descriptive. Studies of program participants report improvements in physical functioning, emotional and mental well-being, mobility, , lower inpatient admissions and self-reported healthcare expen-

ditures(Belza et al. (2006); Hamar et al. (2013) ; Brady et al. (2018) ; Nguyen et al. (2008)). However, these studies examine individuals who are already participating in the programs, raising concerns about systematic differences among individuals in terms of motivation, baseline health, and other determinants of exercise behavior(Newhouse (2014)). More broadly, a related literature uses the age-65 Medicare eligibility threshold to study the effects of Medicare on healthcare utilization, mortality and financial risks, while other literature examine favorable selections into MA plans. Card et al. (2008) and Card et al. (2009) show that Medicare eligibility at age 65 increases healthcare use and improves outcome for seriously ill patients, while Barcellos and Jacobson (2015) show that Medicare reduces exposure to medical expenditure risk. On the Medicare plan choice margin, it has been documented that MA enrollees differ systematically from TM enrollees, underscoring the existence of selections in plan choice (Abaluck and Gruber (2011); Abaluck and Gruber (2016) ; Cabral et al. (2018)). Taken together, these studies underscore both the policy relevance of Medicare at age 65 and the identification challenges regarding MA enrollment, but they leave open question of whether MA is associated with changes in preventive health behavior.

This paper examines whether MA enrollment is associated with a differential change in physical activity at age 65. Using data from the Health and Retirement Study from 2012 to 2022, we estimate a difference-in-discontinuities design, following Grembi et al. (2016)), that compares the age 65 discontinuity in vigorous exercise between MA enrollees and non-MA individuals. The intuition is that even though both MA and non-MA groups experience the universal shocks age age 65 such as retirement, Social Security, general aging, only MA enrollees gain access to structured fitness benefit embedded in MA plans. Under the identifying assumption that the two groups would otherwise experience similar discontinuities at age 65, the difference in discontinuities captures the MA-associated change in exercise behavior at the threshold.

The estimates implies that MA enrollment is associated with an approximately 4 percentage point increase in the probability of daily vigorous exercise at age 65 relative to non-MA groups. Event study confirms no evidence of pre-trends but the effect is concentrated in the short run. The response is larger among females, non-White respondents and individuals without a college degree. A series of robustness checks support our main findings.

Our contribution is threefold. First, we provide new evidence linking MA-specific benefit design to physical activity, which has received very limited attention in the literature regarding preventive health behavior despite growing policy relevance. Further, we document that the behavioral response is concentrated in the short run which speaks to a broader question in behavioral economics about whether financial incentives can promote durable healthy habit formation. Whereas most of the literature on financial incentives and physical activity has focused on younger population, this paper provides the first evidence that similar behavioral responses also emerge among older adults. Lastly, we show that the response is largest among female, non-White , and non-college degree beneficiaries, suggesting that MA covered fitness benefits may reduce structural barriers that constrain physical activity.

2 Data

The main data source is the Health and retirement Study (HRS), a nationally representative survey of older adults in the United States. We use years from 2012 to 2022 providing detailed information on insurance coverage, demographics, labor market status, and self-reported physical activity. As our empirical strategy uses Medicare eligibility threshold at age 65, a key variable in the analysis is continuous age. We therefore construct precise age by using the respondent's birth year and month with the survey year and month. This allows us to implement the difference-in-discontinuities design using age 65 as the running variable.

The primary outcome measures vigorous physical activity based on the HRS question: *"How often do you take part in sports or activities that are vigorous, such as running or jogging, swimming, cycling, aerobics or gym workout, tennis, or digging with a spade or shovel?"*. Respondents report in five levels : "Every day," "More than once a week," "Once a week," "One to three times a month," or "Hardly ever or never".

We construct a binary indicator that equals to 1 if the respondent reports exercising everyday and 0 otherwise. This threshold isolates individuals who consistently engage in vigorous exercise at the highest observable frequency, providing a conservative and policy-relevant measure of workout frequency. Daily vigorous activity represents the group most likely to comply with the federal Physical Activity Guideline. By using behaviorally distinct threshold available, our outcome capture a clean and interpretable measure of high intensity exercise that best aligns with the behavioral mechanisms linked with Medicare Advantage fitness benefits. Further, we complement this binary outcome by using quasi-continuous and parsimonious recoding ("hardly ever or never" as 0; "once a week" as 1; "one to three times a month" as 1; "more than once a week" as 1; and "every day" as 7) for the robustness check to ensure the main specifications are not driven by the imposed binary cutoff.

In the main analysis, we define insurance groups using observed Medicare coverage histories and compare individuals in Medicare Advantage (MA) with those outside MA. The MA group consists of respondents observed in MA during the sample period, while the comparison group includes respondents observed in Traditional Medicare or not yet enrolled in Medicare. Because the COVID-19 pandemic sharply disrupted physical activity opportunities and health behavior in 2020, we exclude that survey year from the baseline analysis and treat it separately in robustness checks. Table 1 reports summary statistics for the resulting estimation sample.

Table 1: Baseline Characteristics by Medicare Advantage Status (Ages 60–70)

	Non-MA N(%)	MA N(%)	Total N(%)
Gender			
Male	3,212 (44.13)	1,888 (39.79)	5,100 (42.42)
Female	4,066 (55.87)	2,857 (60.21)	6,923 (57.58)
Race			
White/Caucasian	5,033 (69.49)	2,789 (59.04)	7,822 (65.36)
Black/African American	1,347 (18.60)	1,331 (28.18)	2,678 (22.38)
Other	863 (11.91)	604 (12.79)	1,467 (12.26)
Hispanic			
Not Hispanic	5,984 (82.40)	3,785 (79.87)	9,769 (81.40)
Hispanic	1,278 (17.60)	954 (20.13)	2,232 (18.60)
Education			
No college degree	4,700 (65.05)	3,441 (73.10)	8,141 (68.23)
College degree or higher	2,525 (34.95)	1,266 (26.90)	3,791 (31.77)
Physical Activity (Vigorous)			
Every day	229 (3.16)	145 (3.07)	374 (3.12)
>1 per week	1,903 (26.23)	1,173 (24.80)	3,076 (25.67)
1 per week	840 (11.58)	569 (12.03)	1,409 (11.76)
1–3 per month	815 (11.23)	510 (10.78)	1,325 (11.06)
Never	3,468 (47.80)	2,332 (49.31)	5,800 (48.40)
Total observations	7,278	4,745	12,023

Notes: Counts are shown with column percentages in parentheses for respondents ages 60–70. The analytic sample includes 12,023 observations. Sample sizes vary slightly across characteristics due to missing values.

3 Empirical Strategy

A central challenge is that Medicare Advantage enrollment is endogenous: individuals who select Medicare Advantage over Traditional Medicare or no Medicare may differ systematically in health preferences, socioeconomic status, underlying health conditions, and baseline physical activity. A naive comparison of mean workout days across insurance groups conflates the effect of MA fitness benefits with selection.

We address this issue with using a regression discontinuity in difference-in-difference (RDDD) design. The intuition is that Medicare eligibility at age 65 generates a sharp exogenous discontinuity in insurance coverage, but a standard RD at this threshold cannot isolate MA-

specific fitness benefits due to multiple shocks contemporaneously happen at age 65 such as retirement, Social Security claiming, changes in healthcare utilization. Our strategy differences out these confounders by comparing the magnitude of age-65 discontinuity in exercise between Medicare Advantage enrollees who gain access to fitness benefits, and non-Medicare Advantage enrollees who do not receive fitness benefits.

Formally, the RDDD estimand is :

$$\begin{aligned}\text{RDDD} = &\Big[E(Workout \mid Post65,\ MA{=}1) - E(Workout \mid Pre65,\ MA{=}1)\Big] \\ &- \Big[E(Workout \mid Post65,\ MA{=}0) - E(Workout \mid Pre65,\ MA{=}0)\Big]. \qquad (1)\end{aligned}$$

This decomposition can be expressed as follows:

$$\begin{aligned}\Delta^{MA} &= \left(Workout^{MA}_{65,\text{post}} - Workout^{MA}_{65,\text{pre}}\right) \\ &= \underbrace{\text{Universal 65 shock}}_{\text{common to all}} + \text{MA-specific selection} + \underbrace{\text{MA fitness benefit}}_{\text{treatment effect}} + \underbrace{\cancel{\text{Time-invariant selection}}}_{\text{fixed traits}}\end{aligned}$$

$$\begin{aligned}\Delta^{\text{non-MA}} &= \left(Workout^{\text{non-MA}}_{65,\text{post}} - Workout^{\text{non-MA}}_{65,\text{pre}}\right) \\ &= \underbrace{\text{Universal 65 shock}}_{\text{common to all}} + \text{Non-MA-specific selection} + \underbrace{\cancel{\text{Time-invariant selection}}}_{\text{fixed traits}}\end{aligned}$$

Time invariant selection is canceled out in within the group difference and taking the difference across the groups eliminates the components common to both groups:

$$\text{RDDD} = \Delta^{MA} - \Delta^{non\text{-}MA} = \text{MA fitness benefit} + (\text{MA-specific selection} - \text{Non-MA-specific selection})$$

Hence, the RDDD estimand removes the universal 65 shock such as retirement or social security and time invariant selections into Medicare Advantage such as fixed individual differences.

3.1 RDDD Specification

We estimate the following equation for our main specification.

$$
\begin{aligned}
Y_{it} = \alpha &+ \beta\,(\text{Post}_{it} \times MA_i) + \gamma_1\,\text{Post}_{it} + \gamma_2\,MA_i \\
&+ f(Age_{it}) + \text{Post}_{it}\,f(Age_{it}) + MA_i\,f(Age_{it}) \\
&+ (\text{Post}_{it} \times MA_i)\,f(Age_{it}) + \boldsymbol{\delta}^\top X_{it} + \mu_i + \tau_t + \varepsilon_{it}. \qquad (2)
\end{aligned}
$$

where Y_{it} denotes vigorous workout days per week for individual i at time t. Post_{it} equals 1 if $Age_{it} \geq 65$. MA_i equals 1 for individuals ever observed in Medicare Advantage. $f(Age_{it})$ is a smooth local-linear function of age X_{it} includes demographic controls such as gender, race , education level. μ_i and τ_t denote individual and year fixed effects. Standard errors are clustered at the individual level.

In addition, to examine the evolution of the treatment effect and test for pre-trends as well as possible selections, we estimate a dynamic RDDD model where b_{it} indexes one-year age bins relative to age 65 ($b_{it} = 0$ for age 65; $b_{it} = -1$ for 64):

$$
\begin{aligned}
Y_{it} = \alpha &+ \sum_{k \neq -1} \beta_k \mathbb{1}\{b_{it} = k\}\,MA_i + \sum_{k \neq -1} \phi_k \mathbb{1}\{b_{it} = k\} \\
&+ g(Age_{it}) + \text{Post}_{it}\,g(Age_{it}) + MA_i\,g(Age_{it}) + (\text{Post}_{it} \times MA_i)\,g(Age_{it}) \\
&+ \boldsymbol{\delta}' X_{it} + \mu_i + \tau_t + \varepsilon_{it}. \qquad (3)
\end{aligned}
$$

The coefficients β_k measure the age-specific difference in outcomes between MA and non-MA individuals relative to $k = -1$ (age 64). Pre-trend validation requires $\beta_k \approx 0$ for $k < 0$. Post-65 β_k trace the timing and persistence of behavioral responses.

4 Results

4.1 Discontinuities in Workout Behaviour

We begin by presenting nonparametric evidence on exercise behavior around the Medicare eligibility threshold. Figure 1 plots linear fit of the monthly binned means of the probability of daily exercise separately for individuals in the Medicare Advantage group and those in the non-MA comparison group. Prior to age 65, MA and non-MA beneficiaries display nearly parallel trends in daily workout likelihood. In addition, exactly at the eligibility threshold, MA enrollees exhibit a discrete increase in daily workout probability, whereas Non-MA enrollees do not. This visual jump anticipates the parametric RDDD estimates and suggests that MA-linked fitness benefits trigger behavioral responses.

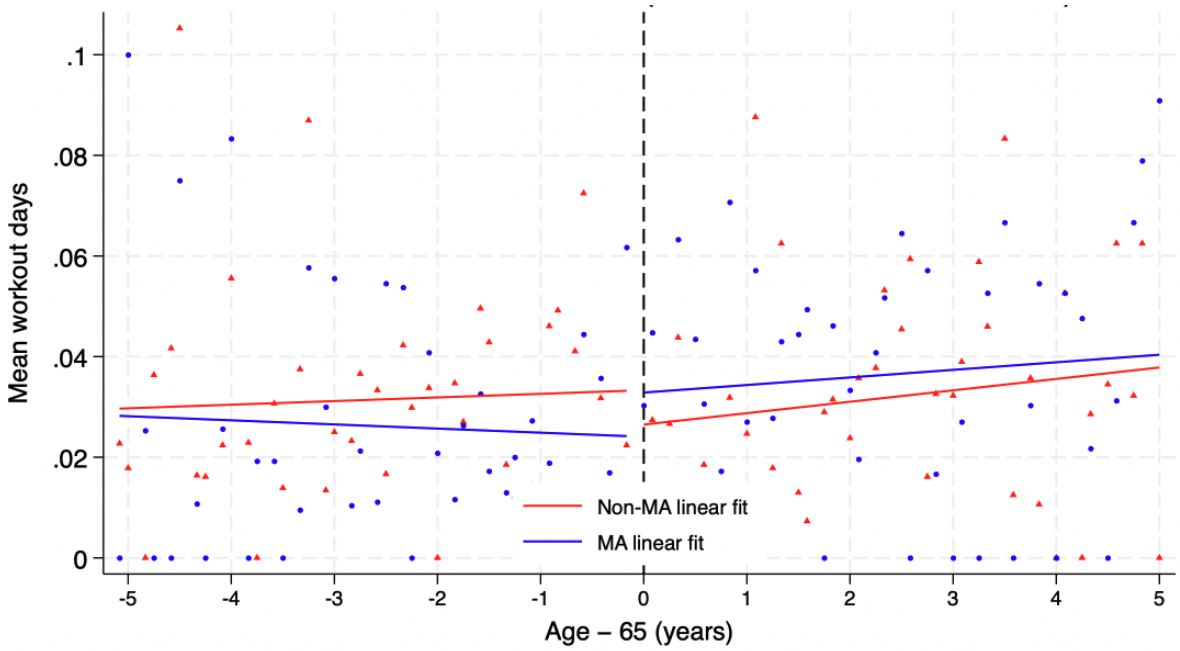


Figure 1
Notes: Binned means of the probability of exercising every day (binary indicator equal to 1 if the respondent reports daily exercise) around age 65 for Medicare Advantage (MA) and non-MA groups. Circles represent MA group means and triangles represent non-MA group means, computed using one-month bins. Solid lines show local linear fits estimated separately on each side of the age-65 cutoff. The vertical dashed line indicates the eligibility threshold at age 65.

Importantly, the figure also demonstrates that while MA enrollees increase their workout

levels at age 65, Non-MA drop theirs. This divergent pattern is consistent with a wide range of age-related health, and lifestyle transitions such as retirement, increased caregiving responsibilities, and broader aging related declines in physical activity. The presence of these countervailing forces at age 65 emphasize why neither a simple RD nor a standard DiD design can isolate the causal effect of MA on physical activity.

A standard RD at age 65 would conflate MA-specific behavioral responses with universal age-65 shocks while a traditional DiD would not effectively control non-linear age profiles or discontinuities common to both groups. The RDDD framework addresses both limitations by explicitly differencing out the age-65 discontinuity experience by both groups, allowing us to attribute the residual discontinuity among MA enrollees specifically to MA fitness incentives rather than to broader aging effect.

4.2 Static RDDD results

Figure 2 presents the first stage discontinuity in Medicare Advantange enrollment at the Medicare eligibility cutoff. Consistent with the institutional setting, Medicare Advantage enrollment increases sharply at age 65, indicating that Medicare eligibility generates substantial variation. This pattern supports the empirical relevance of the RD design and confirms that age 65 provides a meaningful source of identifying exogenous variation.

Table 2 reports reduced-form RDDD estimates of the differential age-65 discontinuity in the probability of daily vigorous exercise for the MA group relative to the non-MA group across several bandwidth choices. Our preferred specification uses a data-driven bandwidth selection procedure following Calonico et al. (2014). Specifically, we adopt the mean squared error optimal bandwidth, denoted by b, as the primary local window around the eligibility cutoff. In our setting, the selected reference bandwidth is approximately 3.5 years. This approach has the advantage of allowing the local estimation window to be determined by

the statistical properties of the data rather than by an arbitrary bandwidth choice, thereby balancing bias and variance in estimation near the threshold.

At this preferred bandwidth, the estimated differential discontinuity implies that MA enrollment increases the probability of daily vigorous exercise by about 4 percentage points relative to non-MA beneficiaries. Although this binary outcome does not directly measure total weekly minutes of exercise, the magnitude is economically meaningful given the low baseline prevalence of daily vigorous activity among older adults. More broadly, the estimate suggests that MA-linked supplemental fitness benefits may induce a measurable increase in preventive health behavior at the point of Medicare eligibility.

To assess sensitivity to the choice of local window, Panel B of Table 2 reports estimates using narrower and wider bandwidths around the cutoff, and Figure **??** plots the corresponding point estimates and 95% confidence intervals across bandwidths. This exercise is particularly relevant in the Health and Retirement Study, which is conducted biennially, since bandwidths of roughly two to four years naturally correspond to comparisons using one or two survey waves on either side of age 65. The figure shows that the estimate obtained using the narrowest window is relatively noisy, reflecting limited precision in a very local sample. In contrast, for bandwidths of 2 years and above, the estimates remain positive and fairly stable, clustering around the main specification. Taken together, the table and figure indicate that the main result is not driven by a single bandwidth choice, but instead reflects a stable positive pattern in the data around the age-65 cutoff.

Figure **??** presents the event study of the RDDD design, which we use primarily for assessing the parallel pre-65 trend which is the critical assumption of our identification strategy. The difference in discontinuity between MA and non-MA beneficiaries are flat and statistically insignificant which supports the key assumption that, in the absence of MA-specific

fitness benefits, both groups would have shown similar discontinuities in the likelihood of daily exercise. At the Medicare eligibility threshold, the event study plot shows a sharp jump in physical activity among MA enrollees relative to non-MA enrollees. The magnitude is considerable and immediate precisely at age 65 which aligns with the activation of MA-linked fitness benefit. However, this effect attenuates and becomes insignificant overtime. This highlights a short-run behavioral boost as individuals gain access to free fitness benefits, consistent with the idea that new financial incentives initially increase the workout frequency.

This attenuation in later years aligns with existing evidence in the physical activity literature among younger population Pojskic et al. (2019), which has shown that financial incentives, fitness subsidies generate short-run and immediate response but it is hard to sustain this positive impact in the long-run. Commitment, habit formation and motivation, rather than financial incentives alone, are the key barriers to long-term adherence in physical activity.

To further address the concern that MA and Non-MA enrollees may already differ in workout level prior to Medicare eligibility, we complement our main RDDD specification with standard event study analysis. Figure 5 presents regular event study comparing raw workout levels (not discontinuity) between MA and non-MA groups. This plot reveals that both groups show similar workout levels and parallel trends prior to Medicare eligibility, with no evidence of systematic divergence in pre-treatment periods. Further, at age 65, we observe a considerable gap with MA beneficiaries increasing their workout frequency while non-MA individuals show no comparable change. However, we do not interpret these post-65 coefficients as causal estimates since enrollment in Medicare Advantage versus enrollment in Traditional Medicare are both endogenous choices.

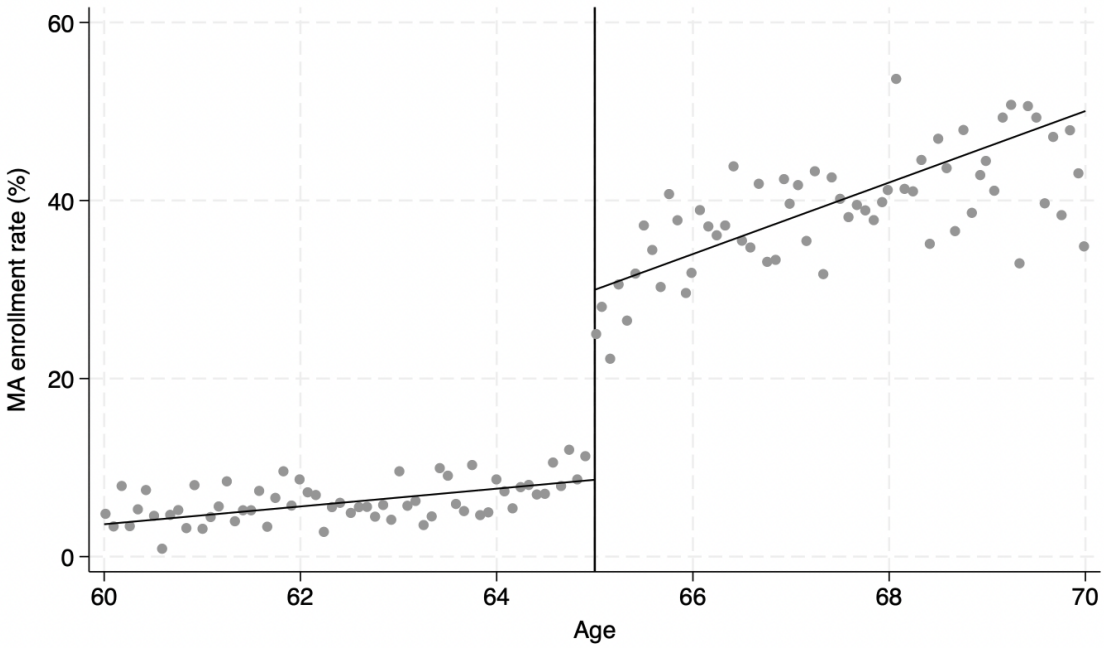


Figure 2
First stage Discontinuity in Medicare Advantage Enrollment at Age 65

Table 2: RDDD Estimates of the Differential Jump in Daily Exercise at Age 65

	Panel A: Main	Panel B: Alternative Bandwidths			
	(1) $h = 3.53$	(2) $h = 1$	(3) $h = 2$	(4) $h = 4$	(5) $h = 5$
$\Delta^{MA} - \Delta^{\text{Non-MA}}$ (pp)[a]	0.040***	0.441*	0.033	0.031**	0.020
	(0.015)	(0.247)	(0.030)	(0.014)	(0.012)
Pre-65 mean, MA group[b]	0.026	0.015	0.022	0.025	0.025
Post-65 mean, MA group[b]	0.037	0.029	0.039	0.038	0.038
Pre-65 mean, Non-MA group[b]	0.031	0.040	0.026	0.031	0.030
Post-65 mean, Non-MA group[b]	0.033	0.024	0.025	0.033	0.033
Observations	8,718	384	3,816	10,009	11,891
R^2	0.498	0.505	0.574	0.476	0.432
Individual fixed effects	✓	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓	✓

Notes: Table reports reduced-form RDDD estimates of the differential discontinuity at age 65 in a binary indicator for daily exercise, using different bandwidth choices h (in years). The dependent variable equals 1 if the respondent reports exercising "every day", and 0 otherwise. The coefficient $\Delta^{MA} - \Delta^{\text{Non-MA}}$ captures the difference between the jump at age 65 for individuals in the MA group and the corresponding jump for the non-MA comparison group.[a] Coefficients are reported in percentage points.[b] Means are computed within the estimation sample for each bandwidth, separately by MA status and by whether age is below (Pre-65) or above (Post-65) the cutoff; standard errors clustered at the individual level are reported in parentheses. All specifications include individual and year fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

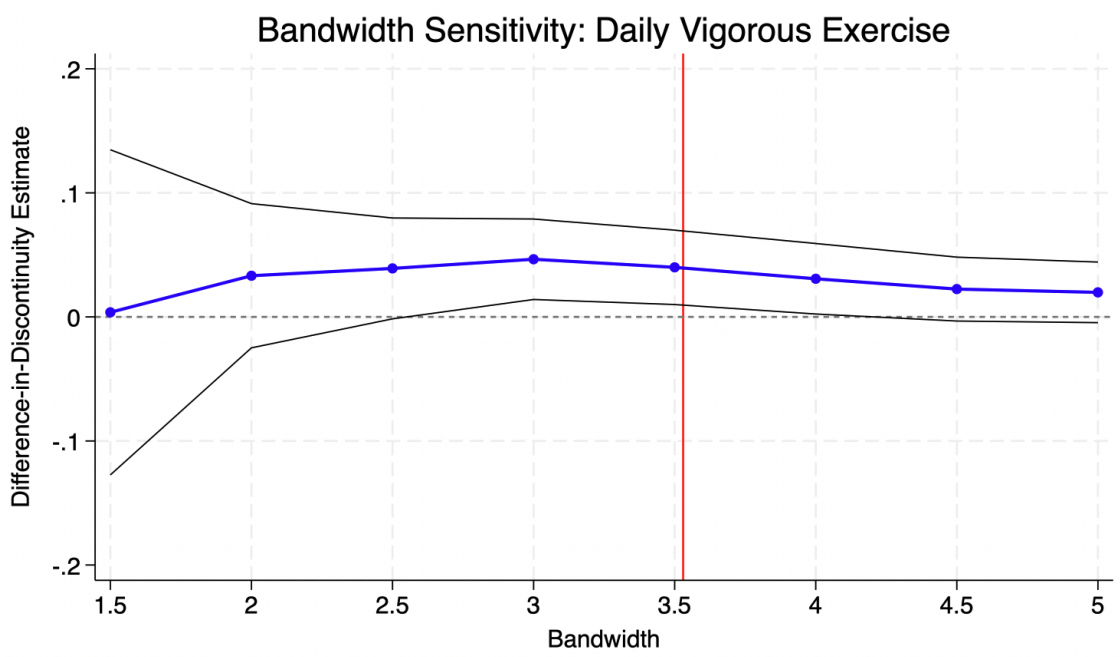


Figure 3
Bandwidth Sensitivity of the Difference-in-Discontinuity Estimate. The figure plots the estimated RDDD coefficient for daily vigorous exercise across alternative bandwidths around the age-65 cutoff. The solid line denotes the point estimate and the surrounding lines denote 95% confidence intervals. The vertical line marks the preferred data-driven bandwidth of 3.53 years.

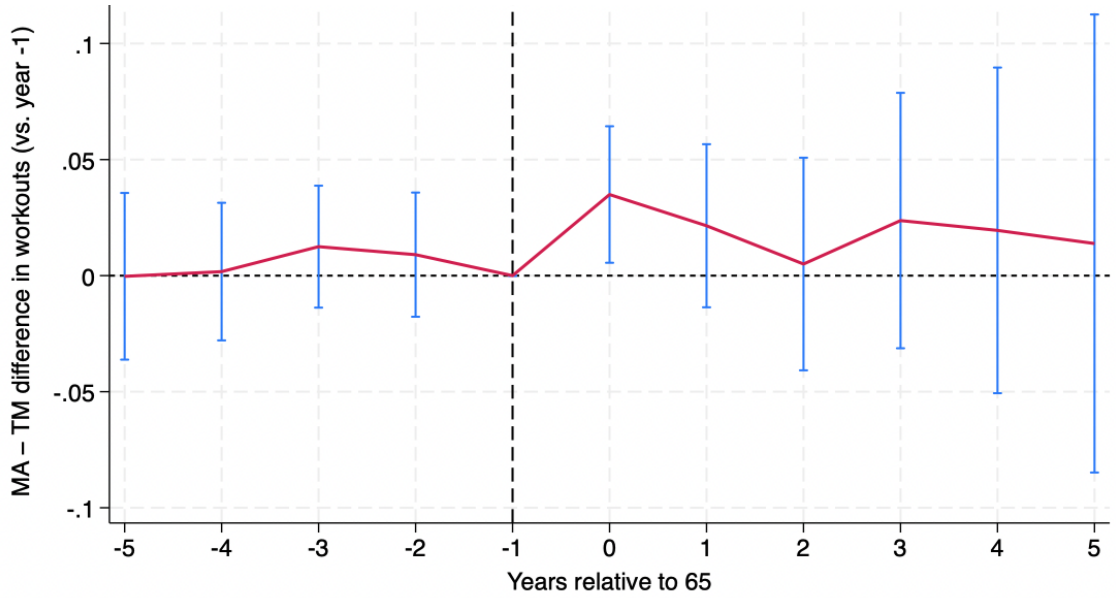


Figure 4
Notes: Event-study estimates of yearly daily exercise probability between Medicare Advantage (MA) and Non-MA beneficiaries in workout days, relative to age 65. The dotted horizontal line indicates zero difference and the dashed vertical line marks a reference year which is a year before Medicare eligibility. Points represent bin means and bars show 95% confidence intervals.

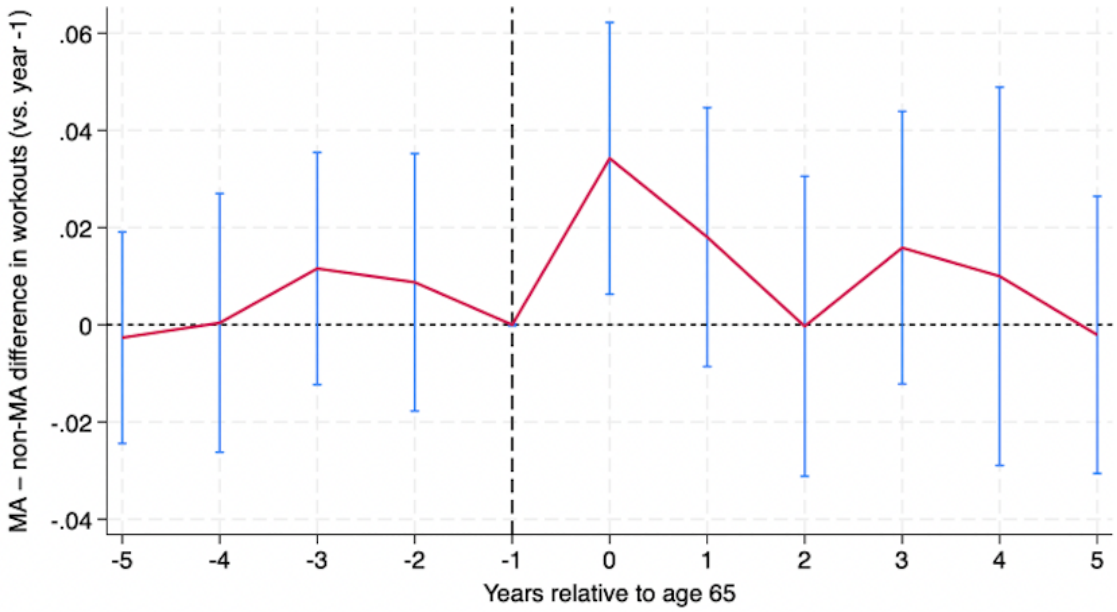


Figure 5
Notes: Event-study estimates of yearly differences in workout level between Medicare Advantage (MA) and Non-MA beneficiaries, relative to a reference year. The dotted horizontal line indicates zero difference and the dashed vertical line marks a reference year which is a year before Medicare eligibility. Points represent bin means and bars show 95% confidence intervals.

5 Heterogeneous Analysis

We investigate whether the RDDD differential effect of medicare eligibility on daily exercise differs across subgroups. Heterogeneity analysis is crucial for our understanding of how behavioral response to Medicare Advantage's fitness benefits. We examine heterogeneity along three dimensions : Gender, Race/Ethnicity (White vs all other groups) , Educational attainment (College degree vs No College degree). For each subgroup analysis, we estimate the same reduced-form RDDD specification used in the main analysis and report static RDDD estimate using data-driven optimal bandwidths and alternative bandwidths as well as dynamic event study.

The heterogeneity results presented in Figure 6 underscore the meaningful variations in behavioral responses to Medicare Advantage (MA) fitness benefits at age 65. Female ben-

eficiaries experience an increase of approximately 5–6 percentage points in the probability of exercising daily, while the corresponding estimate for males is smaller and statistically insignificant. Racial difference is particularly conspicuous where Non-White beneficiaries experience a large increase of approximately 14 percentage points. We do not detect any significant impact among White beneficiaries. Educational differences mirror this pattern as well. College-educated respondents show small and statistically insignificant effects, while non-college educated seniors show statistically significant increases of roughly 5 percentage points. Overall, we find meaningful impact among female, Non-White and non-college educated beneficiaries.

This evidence emphasizes MA supplemental fitness benefits may generate larger gains among populations that historically face greater structural barriers to physical activity which speaks to equitable healthcare access. Non-White and lower-educated seniors often face financaial and informational frictions, limited access to recreational facilities, and higher baseline health risks. The substantial effects among disadvantaged groups is consistent with the interpretation that MA fitness incentives may help reduce disparities in preventive health behaviors at later life. Even though the analysis does not necessarily measure the inequality reduction, the distribution of estimated impacts aligns with an equity enhancing mechanisms

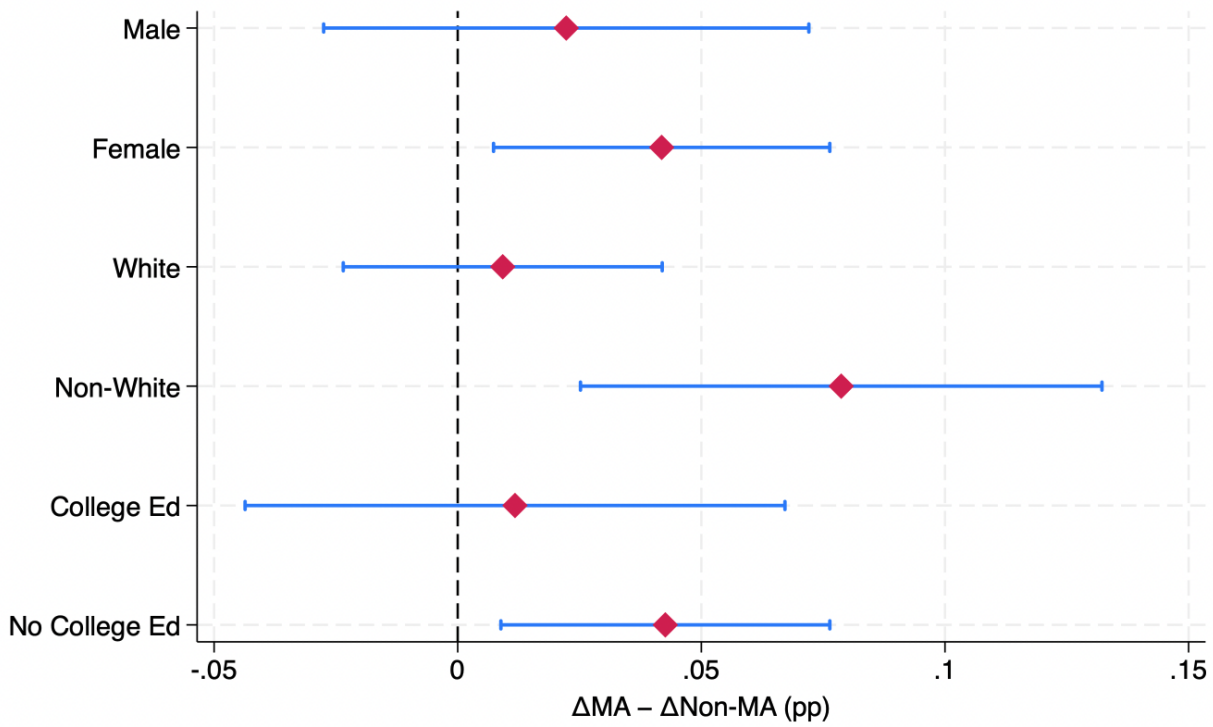


Figure 6
Notes: Reduced-form regression discontinuity in difference-in-discontinuities (RDDD) estimates of the differential jump at age 65 in daily exercise across demographic subgroups. Points denote subgroup-specific estimates of $\Delta^{MA} - \Delta^{\text{Non-MA}}$ expressed in percentage points. Horizontal bars represent 95% confidence intervals based on standard errors clustered at the individual level. Estimates are computed using the MSE-optimal bandwidth and include individual and year fixed effects.

6 Robustness Check

To assess the validity of our results, we perform a comprehensive set of robustness checks that address potential concerns about our findings. These series of robustness checks support the estimated differential discontinuity at age 65 represents a genuine behavioral response associated with the enrollment in Medicare Advantage with fitness benefits, rather than the artifacts of model specification, age related dynamics,and local irregularities in the running variables, or spurious discontinuities.

6.1 Alternative Outcomes: Retirement, Private Insurance Enrollment, Alternative Exercise Measure, and Moderate Physical Activity

We estimate the same RDDD specification using several alternative outcomes that may change around Medicare eligibility and potentially affect workout frequencies: retirement, private insurance coverage, an alternative measure of vigorous exercise, and moderate physical activity. The purpose of these tests is to rule out the possibility that the estimated physical activity jump reflects broader life-cycle transitions around age 65. Consistent with expectations, retirement does not show a meaningful differential discontinuity at age 65, as shown in Table **??**. In addition, we find a statistically significant decline in the number of private insurance plans at age 65 for the MA group relative to the non-MA group, as shown in Table **??**. This sharp reduction indicates that seniors are not simultaneously holding private coverage that could independently encourage exercise. The only insurance margin that changes discontinuously at age 65 for the MA group is the shift into MA plans, which uniquely offer structured free fitness benefits.

We also examine whether the main result is robust to using an alternative measure of vigorous physical activity that preserves more information from the original survey categories. Specifically, we construct an exercise-frequency score equal to 7 for respondents who report vigorous activity "every day," 1 for respondents who report vigorous activity more than once per week, once per week, or one to three times per month, and 0 for respondents who report never engaging in vigorous activity. As shown in Table **??**, the estimated differential discontinuity remains positive under this alternative coding. The estimate is marginally significant under the main bandwidth and statistically significant under the $h = 3$ specification, although the magnitude and precision vary across bandwidths. Because this outcome is coded as an exercise-frequency score rather than a binary indicator, the coefficient should be inter-

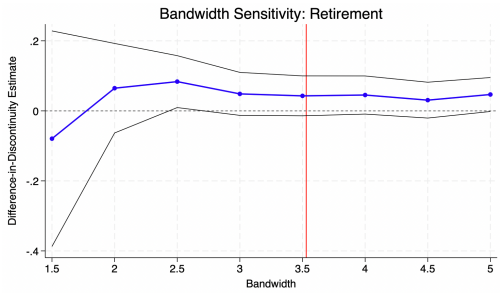
preted in score units rather than percentage points. Overall, this pattern provides additional support that MA enrollment is associated with increased vigorous exercise behavior around age 65.

To assess whether the estimated MA effect reflects an increase in overall physical activity rather than a mechanism linked to structured fitness benefits, we examine a separate outcome measuring daily engagement in moderate physical activities such as gardening, walking at a moderate pace, dancing, and car washing. Table **??** shows that the differential discontinuity at age 65 in this outcome is small and statistically insignificant across all bandwidth choices. The absence of a significant effect on daily moderate activity suggests that our primary findings on vigorous physical activity are unlikely to be driven by broad shifts in activity or general health motivation triggered by Medicare eligibility. Instead, the results are consistent with a more targeted mechanism in which MA-related fitness benefits primarily affect structured or vigorous exercise behaviors, such as gym-based workouts, rather than everyday moderate physical activity.

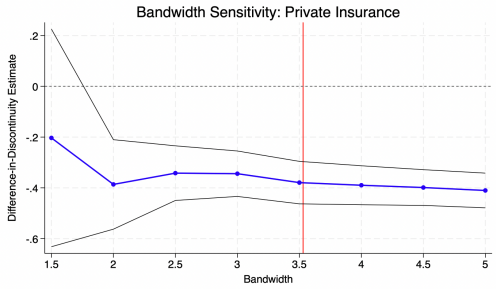
Table 3: RDDD Estimates of Differential Jumps at Age 65 Across Outcomes

	Probability of daily workout			
	Retirement	Private plan	Moderate PA	Alternative workout
	(1)	(2)	(3)	(4)
	$h = 3.530$	$h = 3.530$	$h = 4.541$	$h = 3.820$
$\Delta^{MA} - \Delta^{\text{Non-MAa}}$	0.043	-0.379***	0.019	0.184*
	(0.029)	(0.043)	(0.021)	(0.098)
Mean dependent variable	0.617	0.634	0.095	0.696
Pre-65 mean, MA group[b]	0.566	0.703	0.070	0.658
Post-65 mean, MA group[b]	0.818	0.181	0.099	0.704
Pre-65 mean, Non-MA group[b]	0.499	0.772	0.090	0.710
Post-65 mean, Non-MA group[b]	0.678	0.687	0.122	0.704
Observations	8,556	8,715	11,060	9,418
R^2	0.754	0.688	0.455	0.556
Individual fixed effects	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓

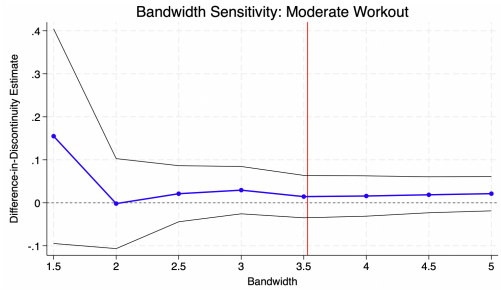
Notes: Columns report RDDD estimates using the preferred bandwidth for each outcome. The dependent variables are retirement status, private plan coverage, daily moderate physical activity, and the alternative workout measure. [a] Means are computed within the estimation sample by MA status and pre-/post-65. Standard errors clustered at the individual level are in parentheses. All specifications include individual and year fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.



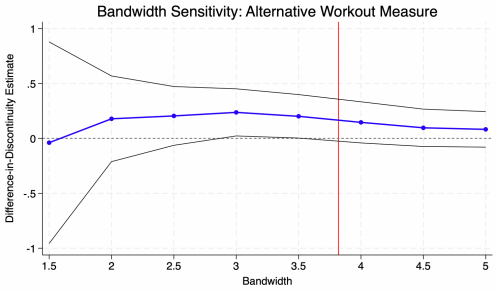
(a) Retirement



(b) Private insurance plans



(c) Daily moderate physical activity



(d) Alternative workout measure

Figure 7
Bandwidth Sensitivity of the RDDD Estimates. The figure plots estimated RDDD coefficients across alternative bandwidths around the age-65 cutoff. The solid line denotes the point estimate, and the surrounding lines denote 95% confidence intervals. The vertical line marks the preferred data-driven bandwidth of 3.53 years.

6.2 Donut hole analysis

Next, we address concerns about the possible manipulation or administrative errors that can happen within a very narrow window around the Medicare eligibility cutoff. To check this local irregularities, we re-estimate the model excluding the window of 3 months pre- and post-65. This "donut hole" deliberately removes the data most sensitive to local misreporting or potential manipulation and tests the stability of our main RDDD specification. As shown in Table4,the RDDD results stay robust and it suggests that our main results are not driven by peculiarities among small group of individuals within the narrow window around the Medicare eligibility cutoff.

Table 4: RDDD Donut-Hole Estimates of the differential jump in workout at age 65

	Panel A: Main	Panel B: Alternative bandwidths			
	(1) $h = 3.53$	(2) $h = 1$	(3) $h = 2$	(4) $h = 4$	(5) $h = 5$
$\Delta^{MA} - \Delta^{\text{Non-MA}}$ (pp)[a]	0.038**	0.441*	0.032	0.029*	0.020
	(0.016)	(0.247)	(0.032)	(0.015)	(0.013)
Pre-65 mean, pooled[b]	0.032	0.029	0.032	0.031	0.031
Pre-65 mean, MA group[b]	0.026	0.015	0.026	0.025	0.025
Post-65 mean, MA group[b]	0.037	0.029	0.037	0.038	0.038
Pre-65 mean, Non-MA group[b]	0.031	0.040	0.031	0.031	0.030
Post-65 mean, Non-MA group[b]	0.034	0.024	0.033	0.033	0.034
Observations	8,611	384	3,717	9,900	11,774
R^2	0.186	-0.031	0.156	0.183	0.173
Individual fixed effects	✓	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓	✓

Notes: Table reports reduced-form RDDD "donut-hole" estimates of the differential discontinuity at age 65 in a binary indicator for daily exercise, using different bandwidth choices h (in years). The donut-hole specification excludes individuals within 3 months on either side of age 65. Means correspond to probabilities of exercising every day. Standard errors clustered at the individual level are reported in parentheses. All specifications include individual and year fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

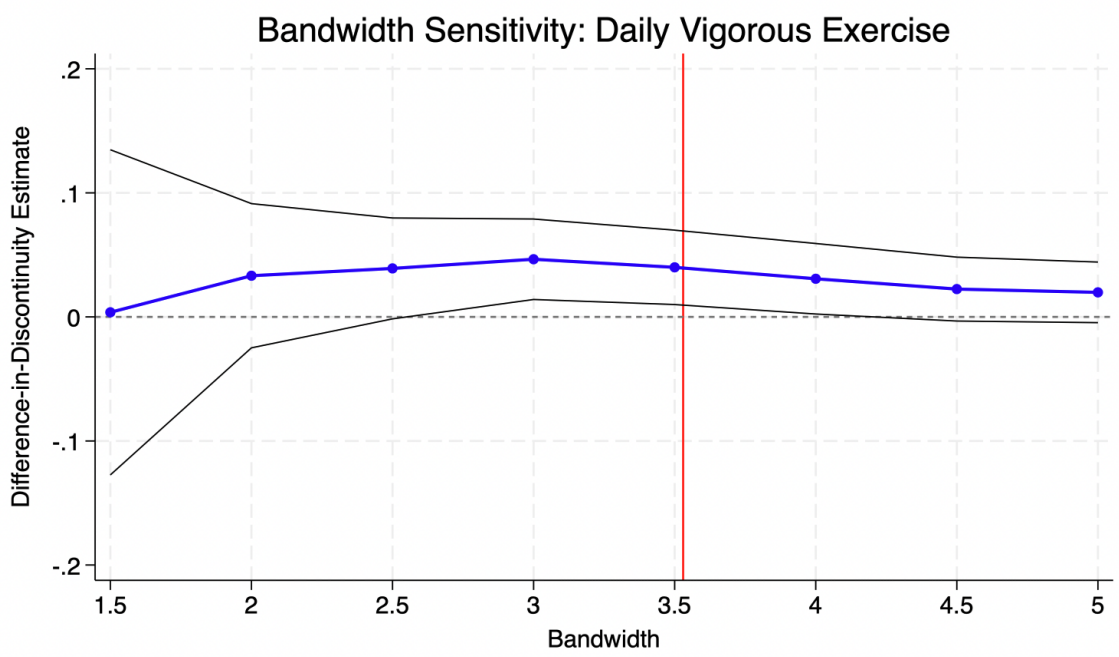


Figure 8
Bandwidth Sensitivity of the Difference-in-Discontinuity Estimate. The figure plots the estimated RDDD coefficient for daily vigorous exercise across alternative bandwidths around the age-65 cutoff. The solid line denotes the point estimate and the surrounding lines denote 95% confidence intervals. The vertical line marks the preferred data-driven bandwidth of 3.53 years.

6.3 Placebo tests at false cutoffs

Lastly, we further validate our empirical design by performing placebo tests at a false cutoff of age 66, which serves as a critical falsification check. Unlike age 65, turning 66 does not correspond to any institutional or policy change in Medicare but possibly correspond to more retirement. Thus, a correctly specified RDDD model should produce no meaningful differential discontinuity at this artificial cutoff. Table5 shows these results and reports the placebo discontinuities statistically insignificant and negative direction. The absence of a spurious jump validates that our main RDDD specification is unique to the true eligibility threshold at age 65.

Table 5: Placebo RDDD Estimates of the differential jump in workout at age 66

	Panel A: Main	Panel B: Alternative bandwidths			
	(1) $h = 4.749$	(2) $h = 1$	(3) $h = 2$	(4) $h = 3$	(5) $h = 5$
$\Delta^{MA} - \Delta^{\text{Non-MA}}$ (pp)[a]	-0.009	-0.126	-0.006	-0.015	-0.018
	(0.014)	(0.155)	(0.036)	(0.021)	(0.014)
Pre-66 mean, MA group[b]	0.027	0.000	0.026	0.026	0.028
Post-66 mean, MA group[b]	0.036	0.048	0.034	0.034	0.038
Pre-66 mean, Non-MA group[b]	0.028	0.000	0.028	0.027	0.029
Post-66 mean, Non-MA group[b]	0.038	0.000	0.035	0.036	0.038
Observations	10,385	116	2,922	6,172	11,036
R^2	0.457	0.547	0.561	0.523	0.444
Individual fixed effects	✓	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓	✓

Notes: Table reports placebo reduced-form RDDD estimates of the differential discontinuity at age 66 (instead of the true age-65 eligibility cutoff) in a binary indicator for regular exercise, using different bandwidth choices h (in years). The dependent variable equals 1 if the respondent reports exercising "every day", and 0 otherwise. The coefficient $\Delta^{MA} - \Delta^{\text{Non-MA}}$ captures the difference between the jump at age 66 for individuals in the MA group and the corresponding jump for the non-MA comparison group.[a] Coefficients are reported in percentage points.[b] Means are computed within the estimation sample for each bandwidth, separately by MA status and by whether age is below (Pre-66) or above (Post-66) the cutoff; values are probabilities of daily exercise. Standard errors clustered at the individual level are reported in parentheses. All specifications include individual and year fixed effects.

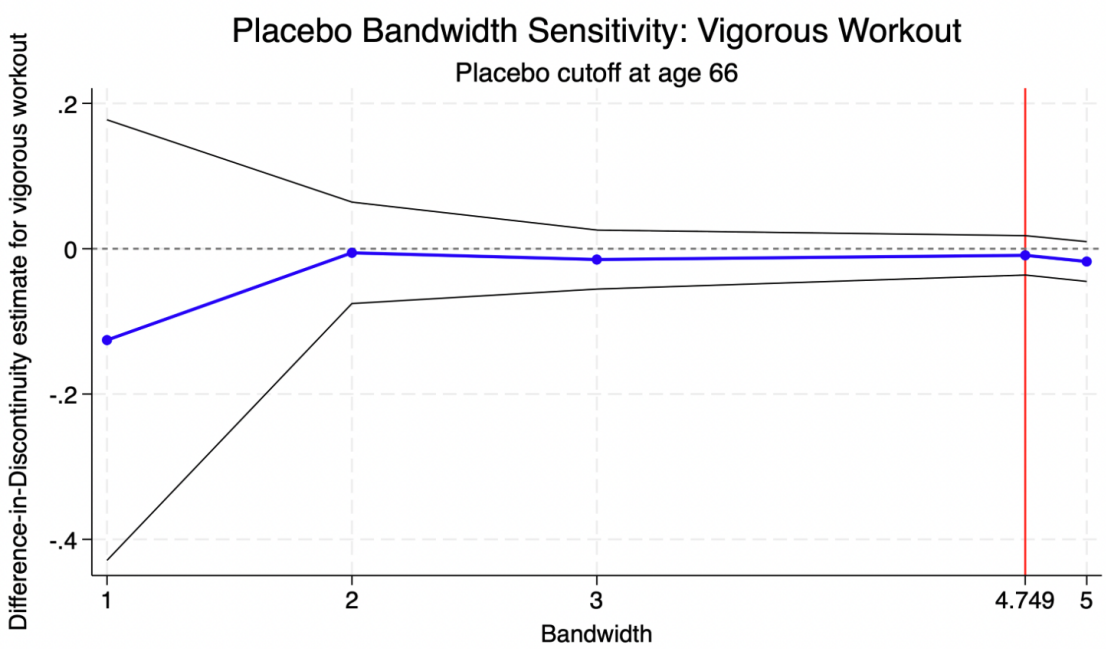


Figure 9
Bandwidth Sensitivity of the Difference-in-Discontinuity Estimate. The figure plots the estimated RDDD coefficient for daily vigorous exercise across alternative bandwidths around the age-65 cutoff. The solid line denotes the point estimate and the surrounding lines denote 95% confidence intervals. The vertical line marks the preferred data-driven bandwidth of 3.53 years.

6.4 Fitness benefit coverage over time

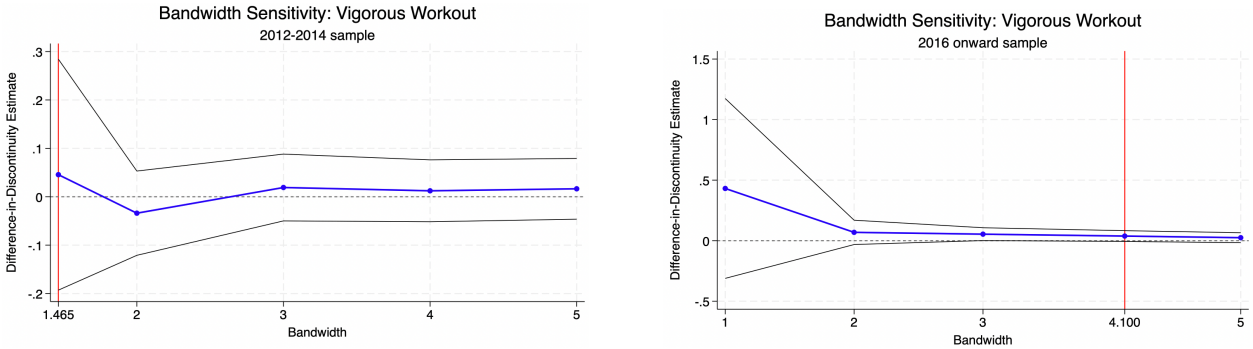
The proportion of Medicare Advantage which covers fitness benefits has expanded substantially during the late 2010s and early 2020s. KFF (2022)reports that in 2022, 97% of individual Medicare Advantage plans offers a fitness benefit, underscoring the near-universal availability of fitness-related supplemental coverage in recent years. We assess whether the impact of MA covered fitness benefit is more pronounced in later years when fitness benefits are more widely available.

For this analysis, we split the analysis into an early period (2012-2014) and a later period (2016–2022). Table **??** shows that estimates in 2012-2014 are small and statistically insignificant across all bandwidth choices, with even negative impact at the optimal bandwidth. In contrast, Table **??** shows more positive, bigger, and statistically significant effects in 2016–2022 for several bandwidths. While the optimal-bandwidth estimate in the later period is not statistically significant, precision is sensitive to bandwidth because the effective estimation sample shrinks substantially at narrow bandwidths. Overall, this earlier year versus later year results are consistent with stronger behavioral responses in the later period when fitness benefits were more broadly available in Medicare Advantage plans.

Table 6: RDDD Estimates of the Differential Jump in Daily Exercise at Age 65 by Period

	Dependent Variable: Daily Exercise	
	2012–2014	2016–2022
	(1)	(2)
	$h = 1.465$	$h = 4.100$
$\Delta^{MA} - \Delta^{\text{Non-MA}}$ a	0.046	0.038*
	(0.122)	(0.023)
Pre-65 mean, MA group	0.009	0.029
Post-65 mean, MA group	0.018	0.042
Pre-65 mean, Non-MA group	0.020	0.034
Post-65 mean, Non-MA group	0.035	0.037
Observations	630	6,045
R^2	0.567	0.538
Individual fixed effects	✓	✓
Year fixed effects	✓	✓

Notes: Columns report reduced-form RDDD estimates using the preferred data-driven bandwidth for each period. Standard errors clustered at the individual level are in parentheses. All specifications include individual and year fixed effects. [a] The coefficient $\Delta^{MA} - \Delta^{\text{Non-MA}}$ captures the difference between the jump at age 65 for individuals in the MA group and the corresponding jump for the non-MA comparison group. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.



(a) 2012–2014 **(b)** 2016 onward

Figure 10
Bandwidth Sensitivity by Time Period. The left panel shows estimates for 2012–2014, while the right panel shows estimates for 2016 onward.

7 Conclusion

We provide the first quasi-experimental evidence on how Medicare Advantage (MA) enrollment shapes seniors' vigorous exercise with the binary indicator for exercising daily. Exercising daily gets higher chance for seniors to meet the federal physical activity guideline which

is more than 150 minutes of moderate-intensity , or 75 minutes vigorous-intensity exercise per week. Exploiting the Regression Discontinuity in Difference in Difference (RDDD), we estimate a reduced-form of differential discontinuity in the probability of exercising everyday at age 65 for MA enrollees compared to Non-MA enrollees. This approach addresses selection concerns by differencing out both observable and unobservable characteristics within and across groups. Using nationally representative panel data from Health and Retirement Study, we find a meaningful and sizable increase in the probability of daily exercise at age 65 for the MA group compared to Non-MA group. These results emphasize the behavioral impact of supplemental fitness benefit which is uniquely provided by MA plans.

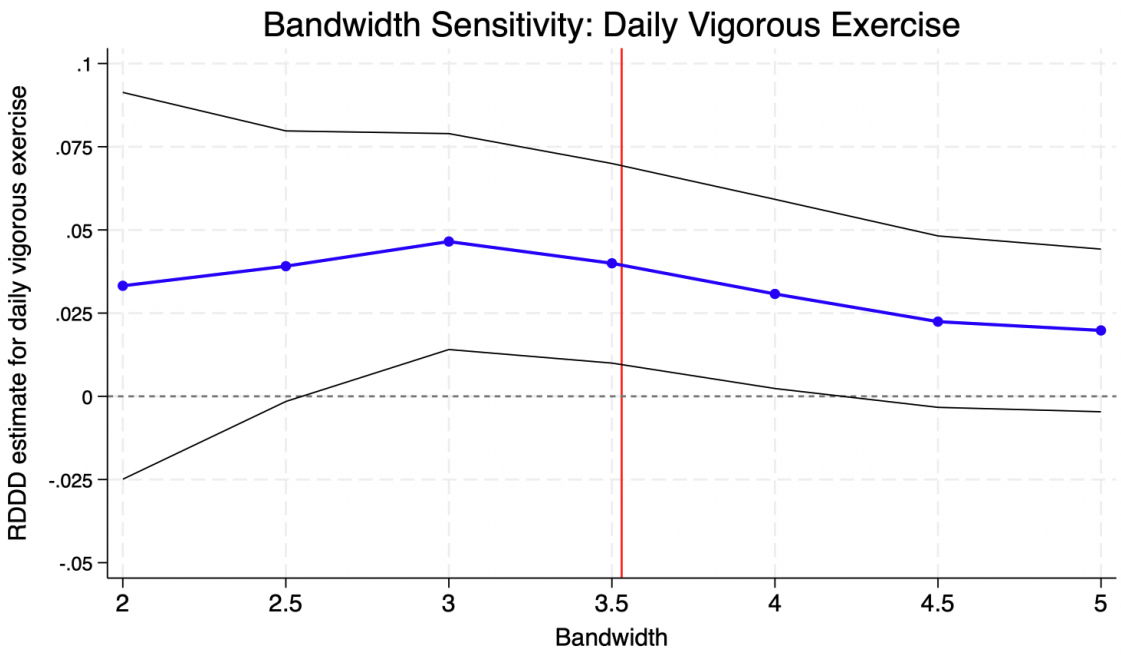
Our findings have several implications and contributions. First, they demonstrate health insurance benefit structure trigger behavioral response, preventive health behavior in particular. By reducing financial and logistical barriers to structured exercise programs, MA plans effectively lead to more active lifestyles when the free fitness benefits activate at the Medicare eligibility. Even modest improvements in vigorous physical activity at older ages may generate substantial effect considering the fact that the baseline physical activity is significantly low among older populations.

In addition, the heterogeneity results underscore who benefits the most from these incentives. The largest increase in daily exercise occur among Non-White senior groups that historically face structural barriers to physical activity and gym access. Our finding signals that fitness benefits embedded in MA plans may be particularly effective in narrowing socioeconomic and racial disparities in healthy aging especially through improving the access to better physical activity.

Lastly, the attenuation of effects after age 65 demonstrates the need for sustained engagement. While MA eligibility induces an immediate behavioral response, maintaining im-

provements in physical activity requires recurring incentives, targeted outreach, government support, or even more personalized wellness program. Our results call attention to the potential value of continued reinforcement rather than one-time eligibility shock.

As MA penetration continues to grow exponentially, the scope and the structure of supplemental fitness benefit will play an increasingly crucial role in shaping health trajectory of the aging population. Our findings emphasize the well-designed, structured insurance programs can function as a policy lever that actively promote healthy aging and provide actionable guidance for policymakers and insurers seeking to design programs that foster healthy life-style.



References

Abaluck, J. and Gruber, J. (2011). Choice inconsistencies among the elderly: Evidence from plan choice in the Medicare Part D program. *American Economic Review*, 101(4):1180–1210.

Abaluck, J. and Gruber, J. (2016). Evolving choice inconsistencies in choice of prescription drug insurance. *American Economic Review*, 106(8):2145–2184.

Barcellos, S. H. and Jacobson, M. (2015). The effects of Medicare on medical expenditure risk and financial strain. *American Economic Journal: Economic Policy*, 7(4):41–70.

Belza, B., Snyder, S., Thompson, M., and LoGerfo, J. P. (2006). The effects of a community-based exercise program on function and health in older adults. *Journal of Aging and Physical Activity*, 14(2):170–182.

Brady, S., D'Ambrosio, L. A., Felts, A., Rula, E. Y., Kell, K. P., and Coughlin, J. F. (2018). Reducing isolation and loneliness through membership in a fitness program for older adults: Implications for health. *Journal of Applied Gerontology*, 39:301–310.

Cabral, M., Geruso, M., and Mahoney, N. (2018). Do larger health insurance subsidies benefit patients or producers? evidence from Medicare Advantage. *American Economic Review*, 108(8):2048–2087.

Calonico, S., Cattaneo, M. D., and Titiunik, R. (2014). Robust nonparametric confidence intervals for regression-discontinuity designs. *Econometrica*, 82(6):2295–2326.

Card, D., Dobkin, C., and Maestas, N. (2008). The impact of nearly universal insurance coverage on health care utilization: Evidence from Medicare. *American Economic Review*, 98(5):2242–2258.

Card, D., Dobkin, C., and Maestas, N. (2009). Does Medicare save lives? *The Quarterly Journal of Economics*, 124(2):597–636.

Carlson, S. A., Fulton, J. E., Pratt, M., Yang, Z., and Adams, E. K. (2015). Inadequate physical activity and health care expenditures in the united states. *Progress in Cardiovascular Diseases*, 57(4):315–323. Estimated that about 8.7

CDC (2022). No leisure-time physical activity among adults — united states, 2021. `https://www.cdc.gov/cdi/indicator-definitions/npao.html`. Accessed: 2025-09-03.

Cubanski, J. and Neuman, T. (2023). What to know about medicare spending and financing. `https://www.kff.org/medicare/issue-brief/what-to-know-about-medicare-spending-and-financing/`. Accessed: 2025-09-03.

Elgaddal, N. and Kramarow, E. A. (2024). Characteristics of older adults who met federal physical activity guidelines for americans: United states, 2022. Technical Report NCHS National Health Statistics Report No. 215, National Center for Health Statistics, Centers for Disease Control and Prevention. Accessed: 2025-09-03.

Fries, J. F., Bruce, B., and Chakravarty, E. (2011). Compression of morbidity 1980–2011: A focused review of paradigms and progress. *Journal of Aging Research*, 2011:Article ID 261702.

Grembi, V., Nannicini, T., and Troiano, U. (2016). Do fiscal rules matter? *American Economic Journal: Applied Economics*, 8(3):1–30.

Hamar, B., Coberley, C. R., Pope, J. E., and Rula, E. Y. (2013). Impact of a senior fitness program on measures of physical and emotional health and functioning. *Population Health Management*, 16(6):364–372.

Jiang, Y. S. and Wang, e. a. (2022). The association between sedentary behavior and falls in older adults: A systematic review and meta-analysis. *Frontiers in Public Health*.

KFF (2022). Medicare advantage 2022 spotlight: First look at plan offerings (slide summa-

rizing supplemental benefits). Reports that 94% of individual Medicare Advantage plans offer a fitness benefit in 2022.

KFF (2025). Medicare advantage in 2025: Enrollment update and key trends. https://www.kff.org/medicare/medicare-advantage-enrollment-update-and-key-trends/. In 2025, more than half (54

Lee, I.-M., Shiroma, E. J., Lobelo, F., Puska, P., Blair, S. N., and Katzmarzyk, P. T. (2012). Effect of physical inactivity on major non-communicable diseases worldwide: an analysis of burden of disease and life expectancy. *The Lancet*, 380(9838):219–229.

Newhouse, J. P. (2014). How successful is medicare advantage? *The Milbank Quarterly*. Analyzes mechanisms—such as enhanced risk adjustment and lock-in rules—implemented to address favorable selection into Medicare Advantage.

Nguyen, H. Q., Maciejewski, M. L., Gao, S., Lin, E., Williams, B., and LoGerfo, J. P. (2008). Health care use and costs associated with use of a health club membership benefit in older adults with diabetes. *Diabetes Care*, 31(8):1562–1567.

Pojskic, H., Sit, C. H. P., Wong, S. H. S., and Sum, R. K. W. (2019). Financial incentives for exercise in older adults: Systematic review and meta-analysis. *Journal of Medical Internet Research*, 21(8):e12098.

U.S. Census Bureau (2018). Older people projected to outnumber children. https://www.census.gov/newsroom/press-releases/2018/cb18-41-population-projections.html. Accessed: 2025-09-03.

Warburton, D. E. R., Nicol, C. W., and Bredin, S. S. D. (2006). Health benefits of physical activity: the evidence. *Canadian Medical Association Journal*, 174(6):801–809.